PROJECT TOPIC: Stock Market Sentiments Analyzer

Programme: MCA

Project Group Members:

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Objective: The stock market is highly influenced by public sentiment, news, and social media trends. Traditional methods of analyzing stocks rely heavily on historical price patterns and technical indicators, which often fail to capture the real-time mood of the market. To address this challenge, we propose a Stock Market Sentiment Analysis and Buy/Sell Recommendation System that leverages Natural Language Processing (NLP) and Machine Learning techniques to analyze financial news, articles, and social media posts. the system collects real-time data, classifies sentiments as positive, negative, or neutral, and generates actionable insights in the form of Buy, Sell, or Hold recommendations. The backend is powered by Spring Boot for scalable API management, while sentiment classification is achieved using Python-based ML models. A user friendly dashboard built with React.js and Chart.js visualizes sentiment trends, price movements, and recommendations. This project demonstrates how AI-driven decision-making can assist investors in reducing human bias, improving trading efficiency, and making data-driven financial decisions. In the future, the system can be expanded to include cryptocurrency analysis, predictive modeling, and automated trading bots for end-to-end smart investment solutions.

Tools required:

- **Hardware Requirements:**

- Processor: Intel i3 or higher
- RAM: Minimum 4 GB
- Storage: At least 10 GB free disk space
- Display: Minimum 1366×768 resolution

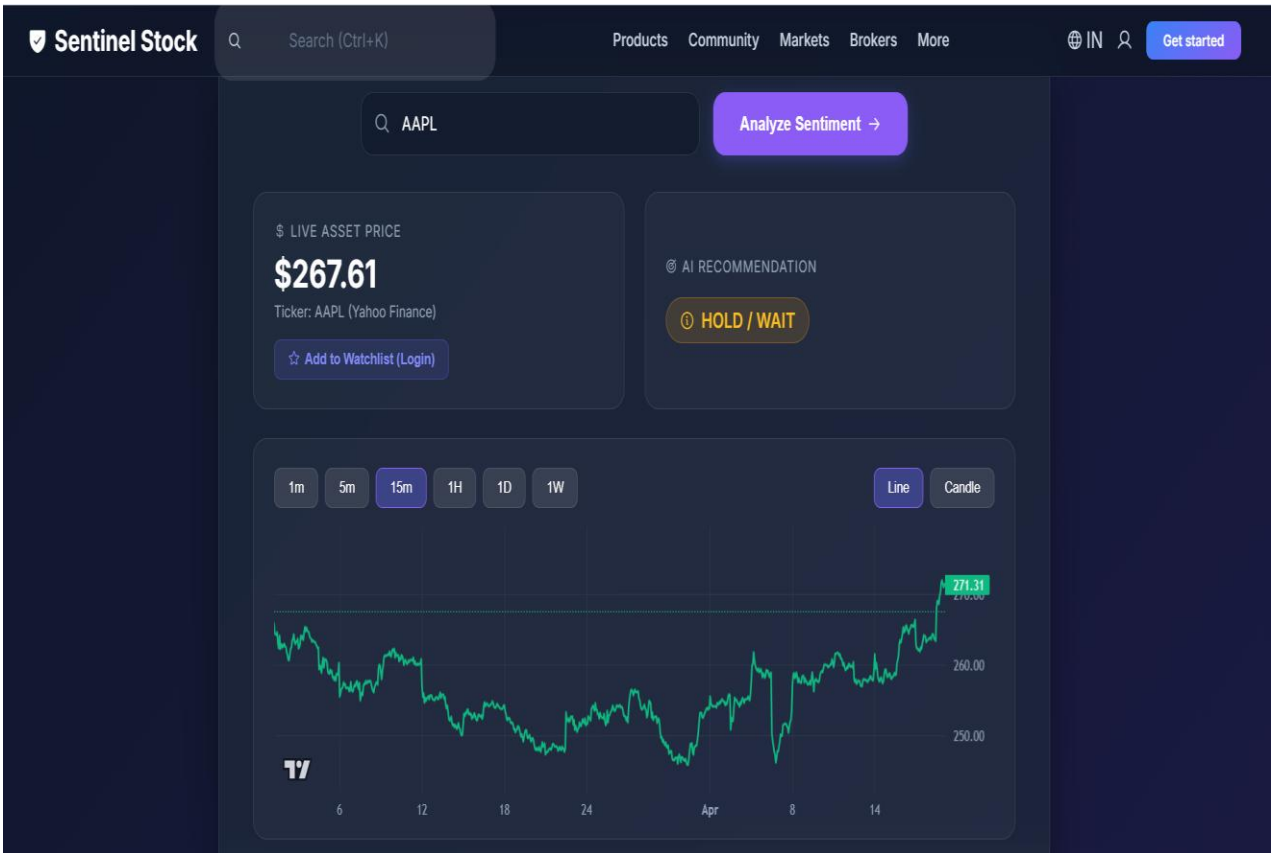
- **Software Requirements:**

- Operating System: Windows 10/11, Linux, or macOS
- Frontend : HTML/CSS/JAVA SCRIPT /AJAX
- Backend : SpringBoot,Python,Mysql
- IDE: VS Code

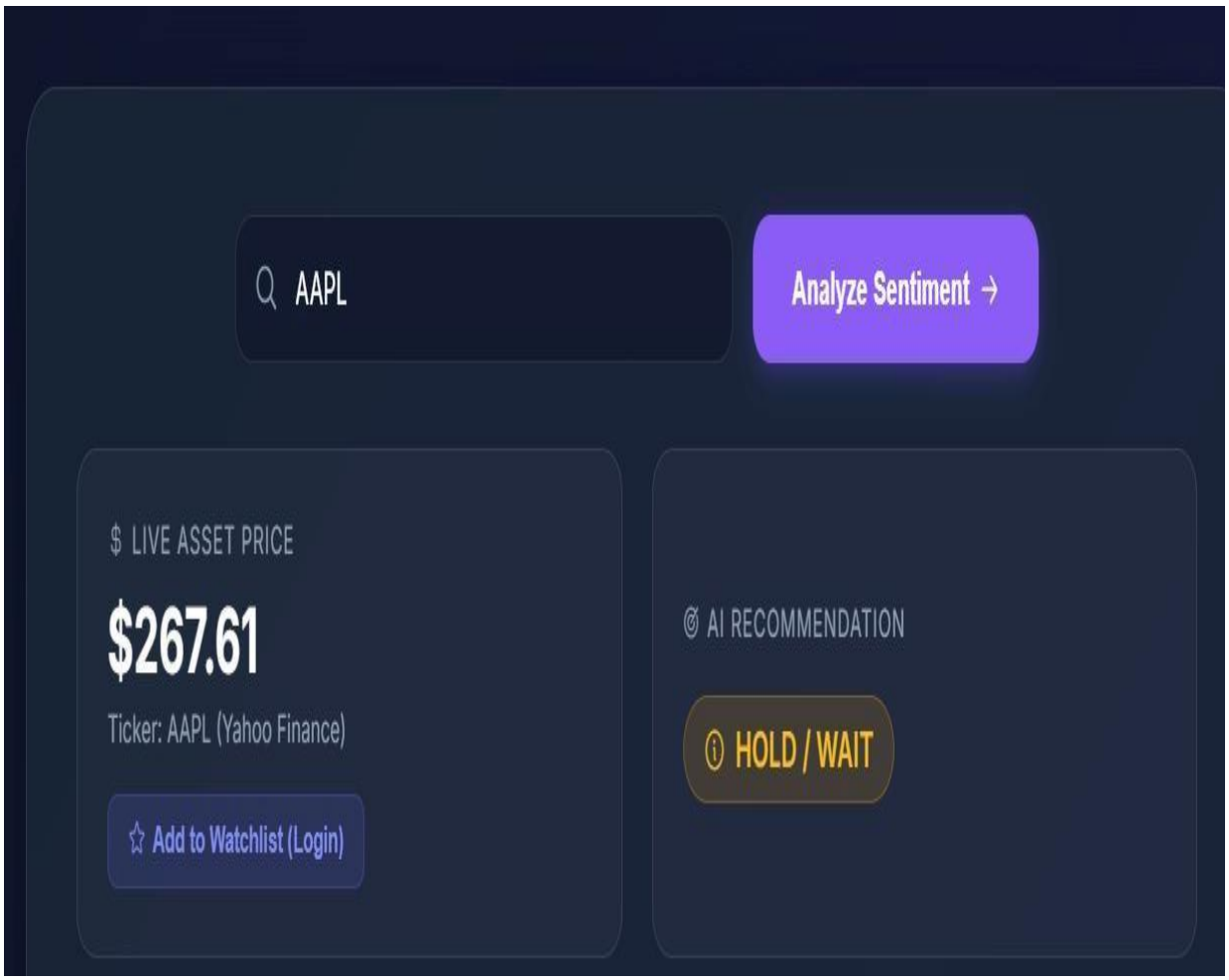
Abstract: The Stock Market Sentiment Analyzer is an AI-driven system designed to assist Indian retail investors by transforming unstructured financial news into actionable sentiment insights. As market movements are heavily influenced by real-time news, social media discussions, and global events, manually interpreting this information often leads to delays and emotional decision-making. This project aims to automate the sentiment analysis process using transformer-based models such as FinBERT to classify financial text into Positive, Negative, or Neutral categories. The system continuously fetches live news related to Indian stock tickers, processes it using NLP techniques, and presents the results through an interactive dashboard featuring sentiment scores, trend visualizations, and historical logs. By providing real-time, data-backed sentiment insights, the platform bridges the gap between retail and institutional investors, enabling more informed and unbiased trading decisions. This project ultimately demonstrates how AI can simplify financial analysis and enhance decision-making efficiency in India's growing stock market.

Home Dashboard Interface

Stock Market Sentiment Analyzer:

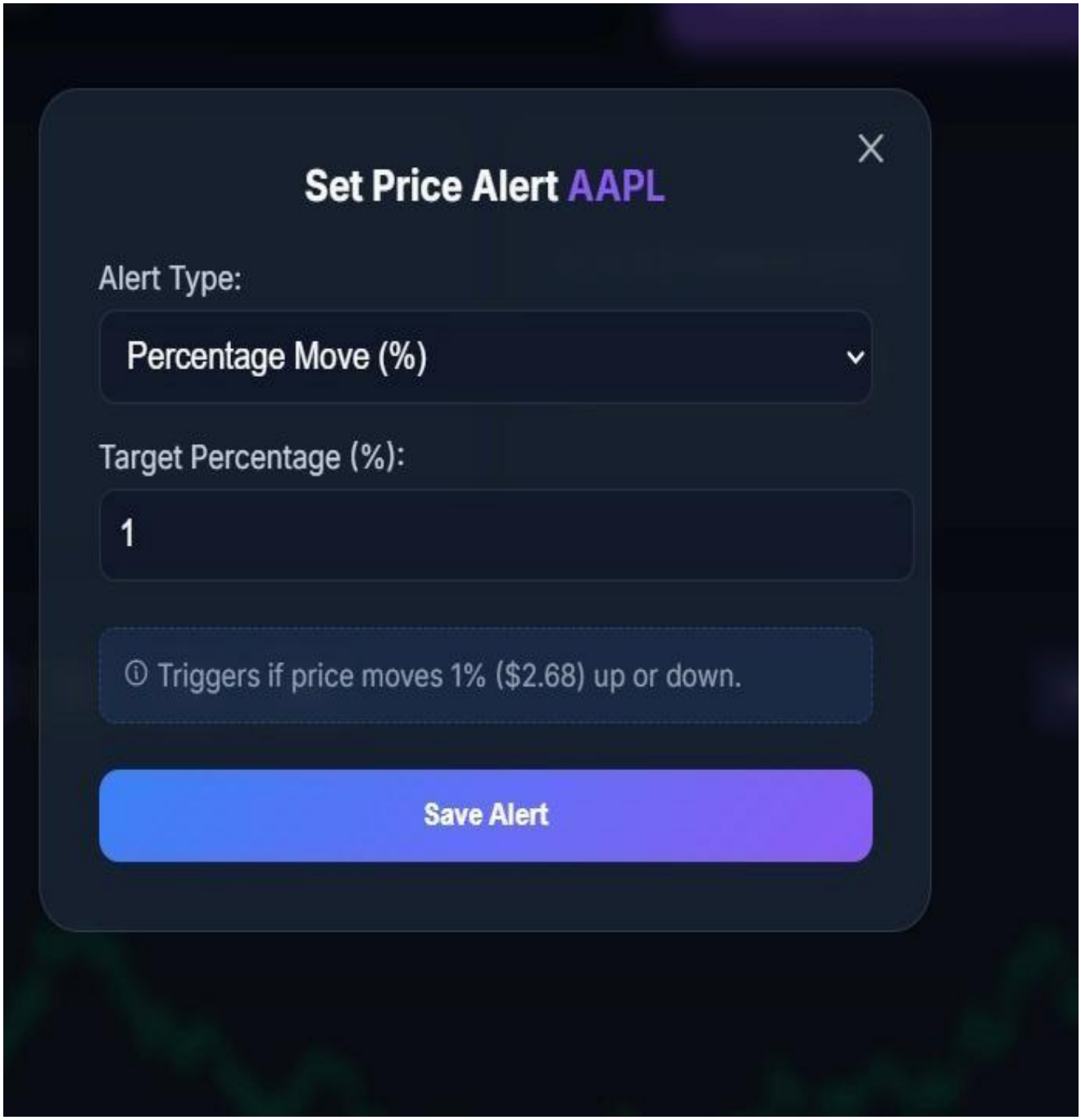


Live Asset Price Card:

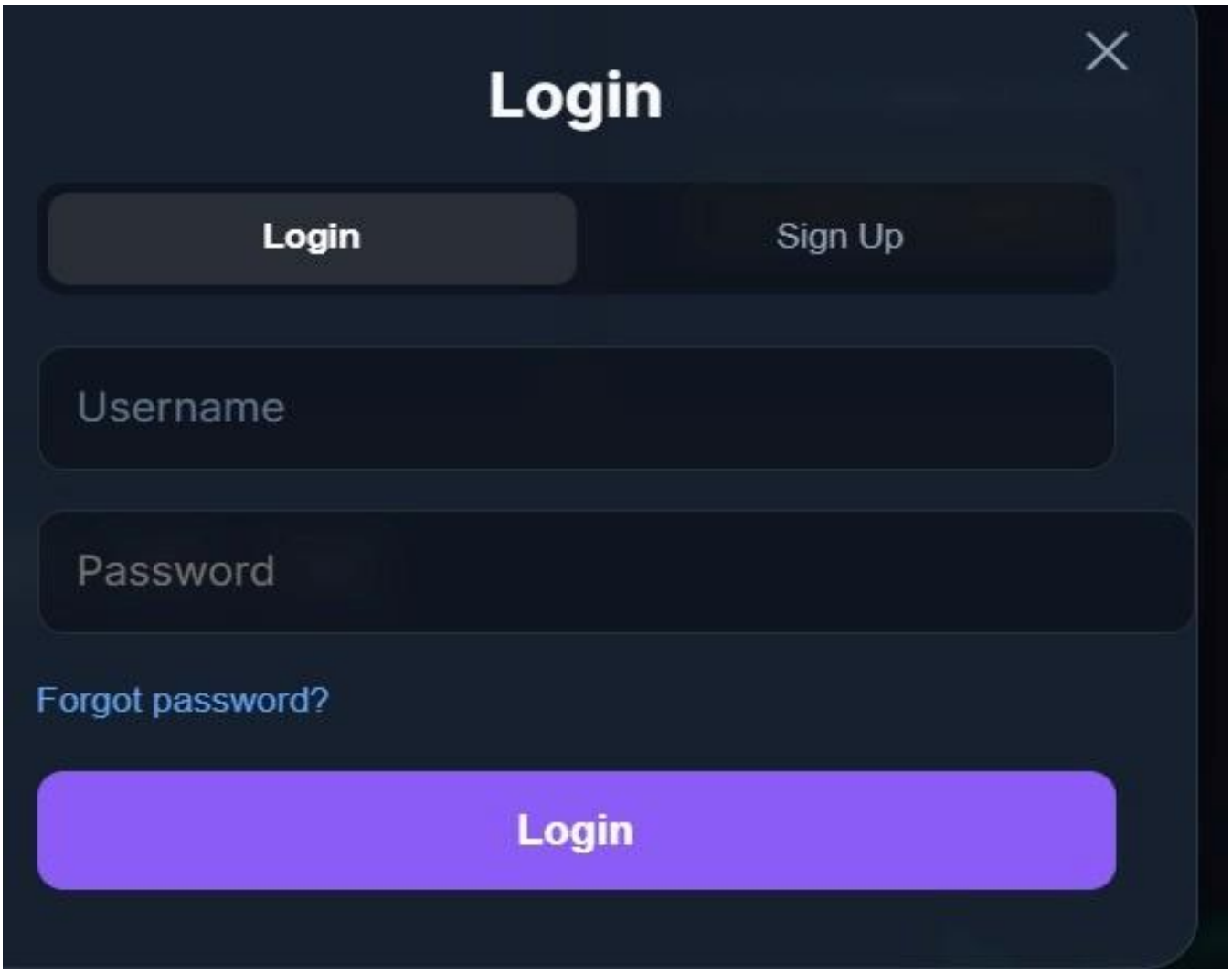


Interactive Stock Chart Module:

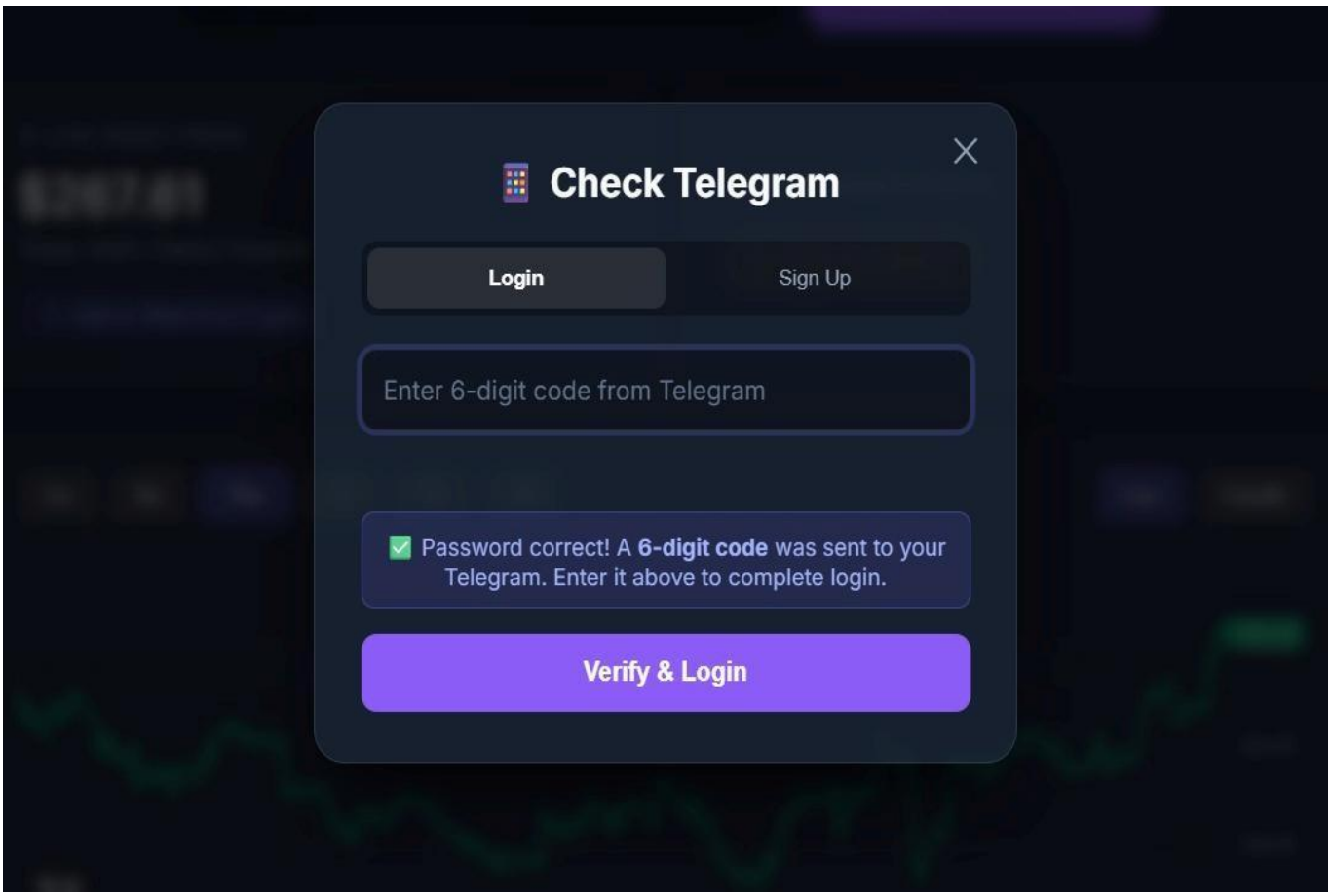
Price Alert Notification System:



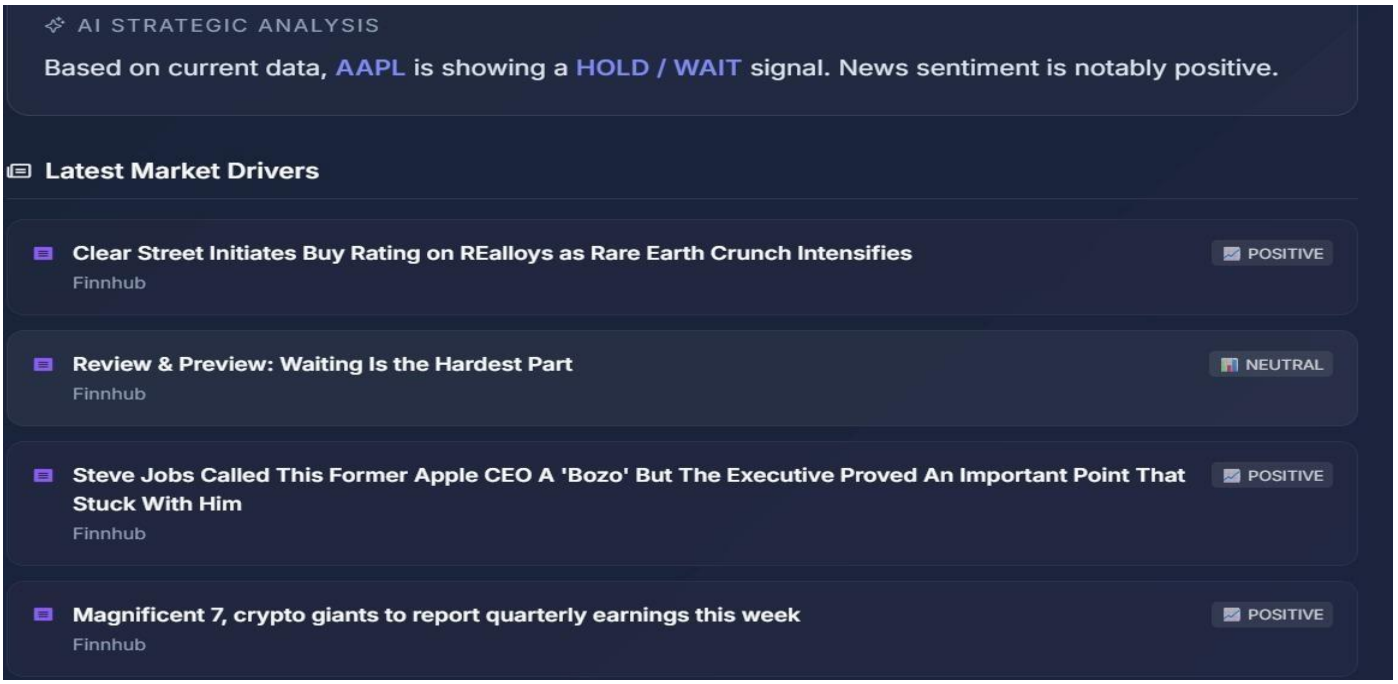
Login and Authentication Screen:



Telegram OTP Verification Integration:



Latest Market Drivers Section:



Sentiment Pie Chart Visualization: